

Wykaż, że jeżeli $a = \log_3 18$, $b = \log_{\sqrt{3}} 12$ to $\log_2 27 = \frac{12}{b-2}$.

① dane:

$$a = \log_3 18, b = \log_{\sqrt{3}} 12 \quad \text{teza: } \log_2 27 = \frac{12}{b-2}$$

korzystamy z własności: $\log_b c = \frac{\log_a c}{\log_a b}$ ①

② upraszczamy najpierw logarytmy a i b

$$a = \log_3 18 = \log_3 (2 \cdot 3^2) = \log_3 2 + \log_3 3^2 = \log_3 2 + 2 \Rightarrow \log_3 2 = a - 2 \quad ②$$

$$b = \log_{\sqrt{3}} 12 \stackrel{①}{=} \frac{\log_3 12}{\log_3 3^{\frac{1}{2}}} = \frac{\log_3 (3 \cdot 2^2)}{\frac{1}{2}} = 2 \log_3 3 + 2 \log_3 2^2 = 2 + 4 \log_3 2$$

② podstawiamy do b :

$$b = 2 + 4(a - 2)$$

$$\frac{1}{4}(b - 2) = a - 2 \quad ③$$

③ upraszczamy $\log_2 27$

$$\log_2 27 \stackrel{①}{=} \frac{\log_3 27}{\log_3 2} = \frac{3}{\log_3 2} \stackrel{②}{=} \frac{3}{a-2} \stackrel{③}{=} \frac{3}{\frac{1}{4}(b-2)} = \frac{12}{b-2} \quad \square$$